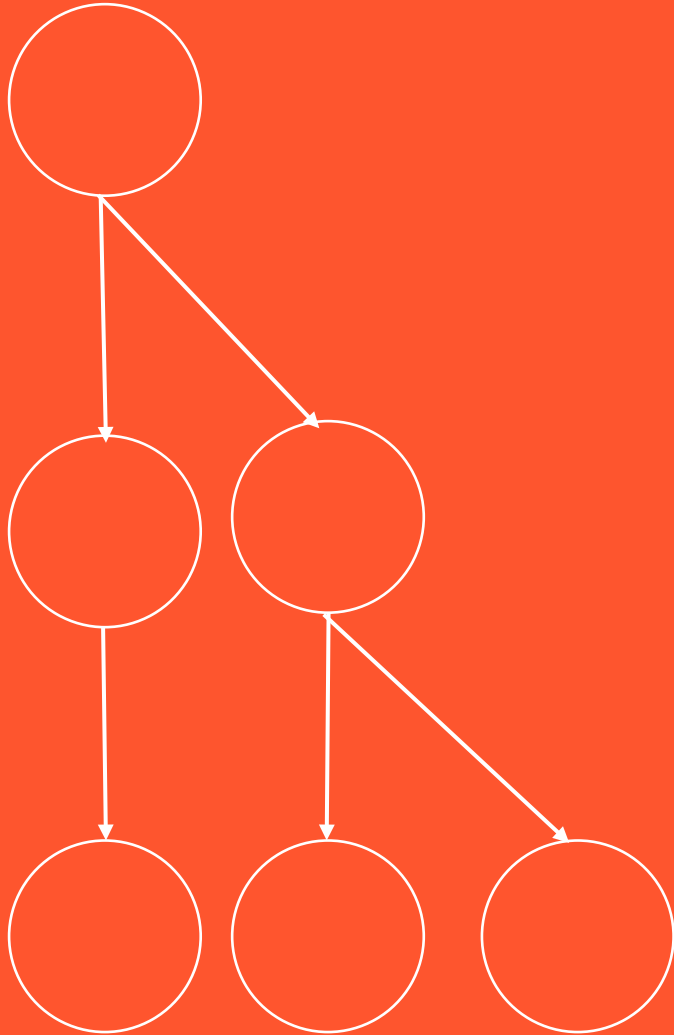


BCOG 100

Social Cognition

Lecture 2: Primate Cognition: Algorithms and Representations



UNIVERSITY OF
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URBANA-CHAMPAIGN

Reading Check 10-2

Password: games

Question: you have repeated interactions with someone. When they do something nice, you are nice back, and when they are mean, you do something mean back. Do you need to represent their mind to behave this way? Why or why not?

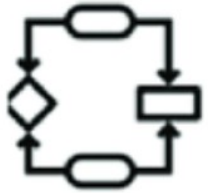


Primate Cognition: Algorithms & Representations



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Last Time:

Last time we identified new primate behaviors

Today:

Today we ask what internal capacities made them possible

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The Changes to Explain

1. Cooperation beyond kin
2. Stable social hierarchies
3. Ingroup-outgroup dynamics
4. Complex internal emotions
5. Feelings of fairness and norms
6. Long-horizon planning
7. Cultural transmission
8. Tool use and mediated skill acquisition
9. Much bigger brains



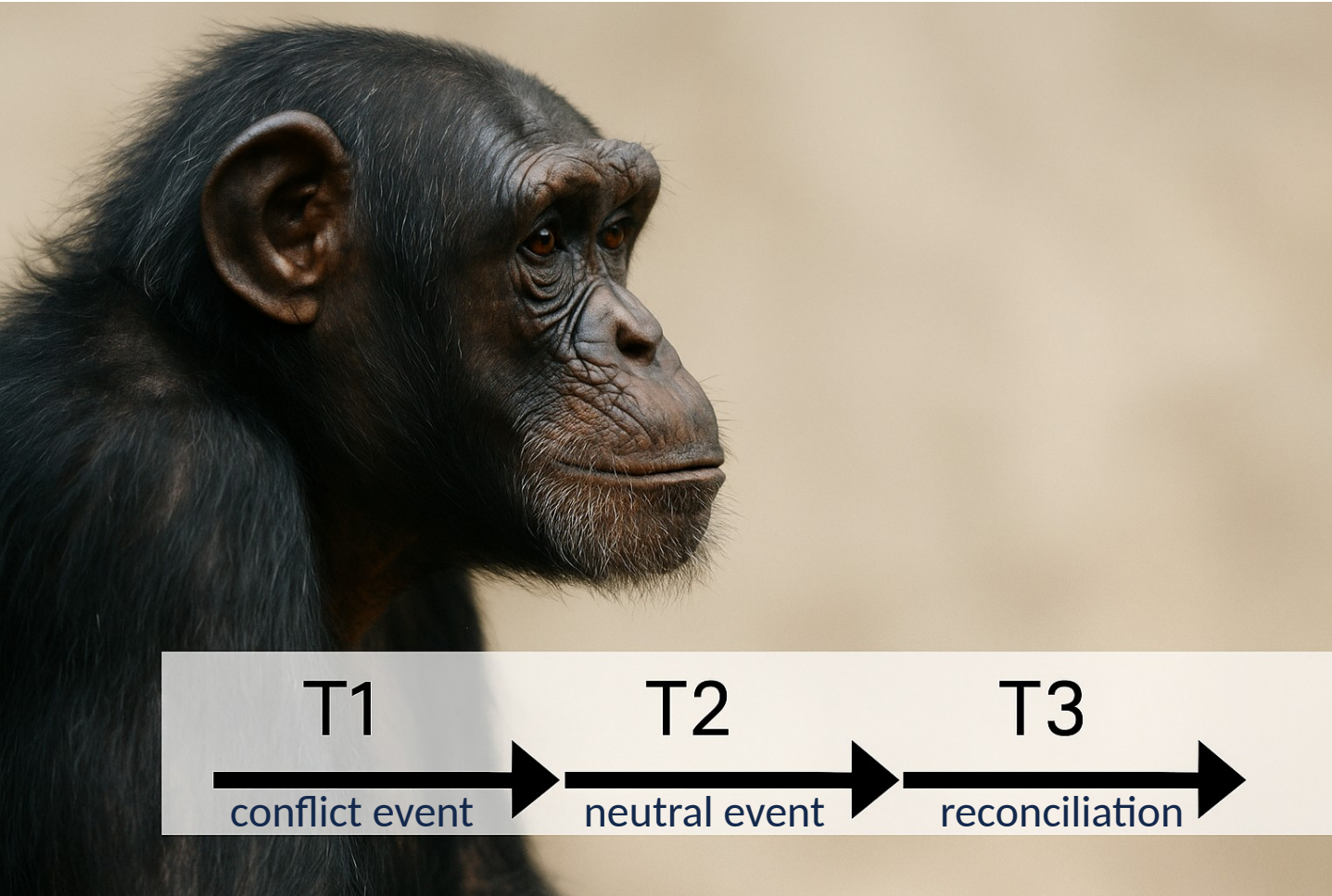
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Why Basic Mammalian Computation Is Not Enough

- Reflex learning and reinforcement learning already existed
- Generative models for sensory prediction already existed
- These could not by themselves yield the primate behavioral phenotype

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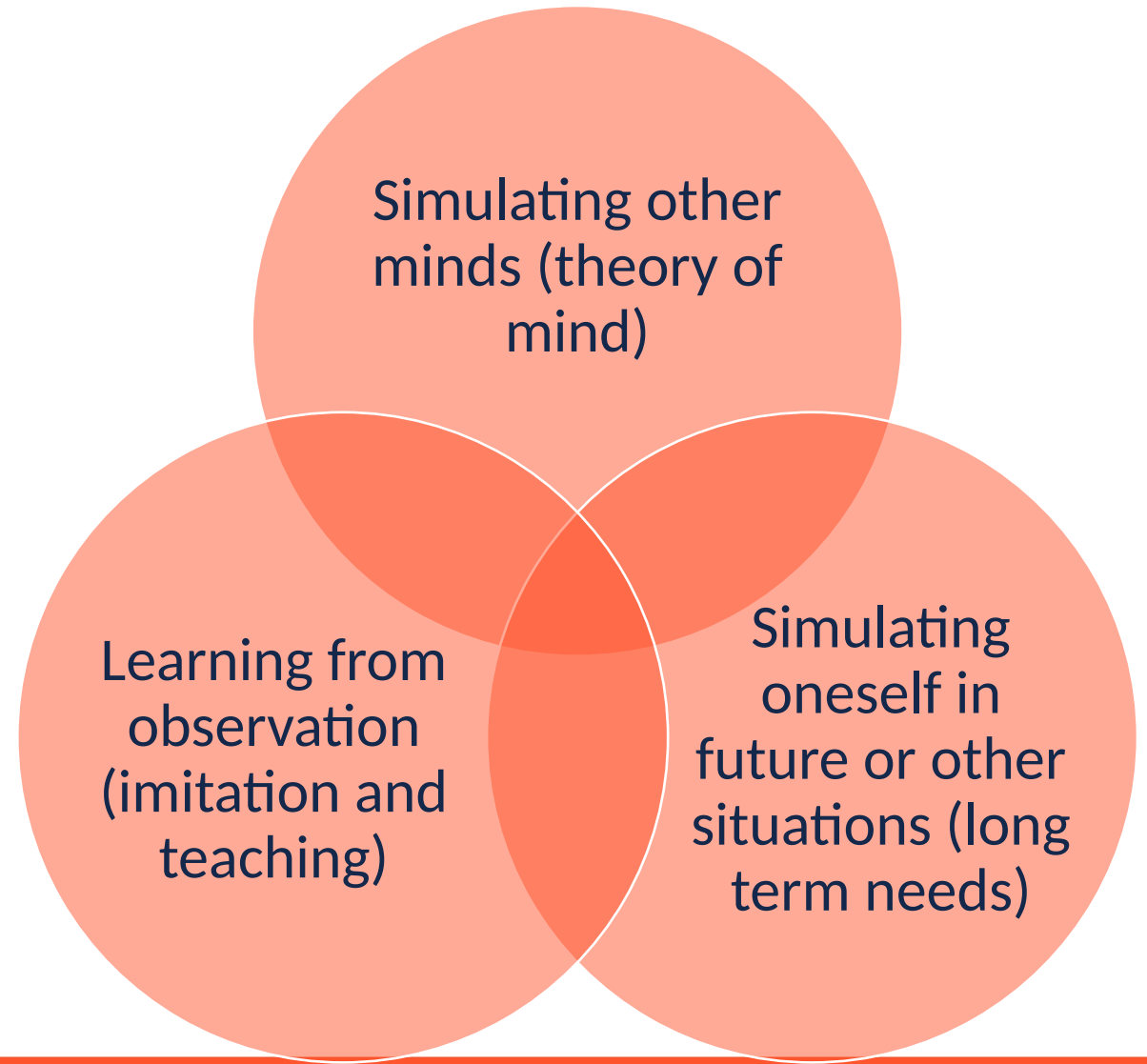


Inference: Internal Representations Carried Forward

- The behaviors imply long-range internal models
- These models referenced futures and other agents
- They altered behavior even when no external cues were present

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Primate Generative Model Capacities



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		Player A	
Player B	Cooperate	Mutual Cooperation	A takes advantage of B
	Defect	B takes advantage of A	Mutual Defection

Cooperation Beyond Kin Required Modeling Others

- Non-kin cooperation depended on predicted reciprocity
- Individuals acted as if future responses were represented
- Social behavior reflected forecasts, not reflexes

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Hierarchy Presupposed Internalizing Others' Dispositions

- Dominance relations guided action across contexts
- Individuals deferred or challenged based on stored expectations
- Rank functioned as an internal variable used in decision-making

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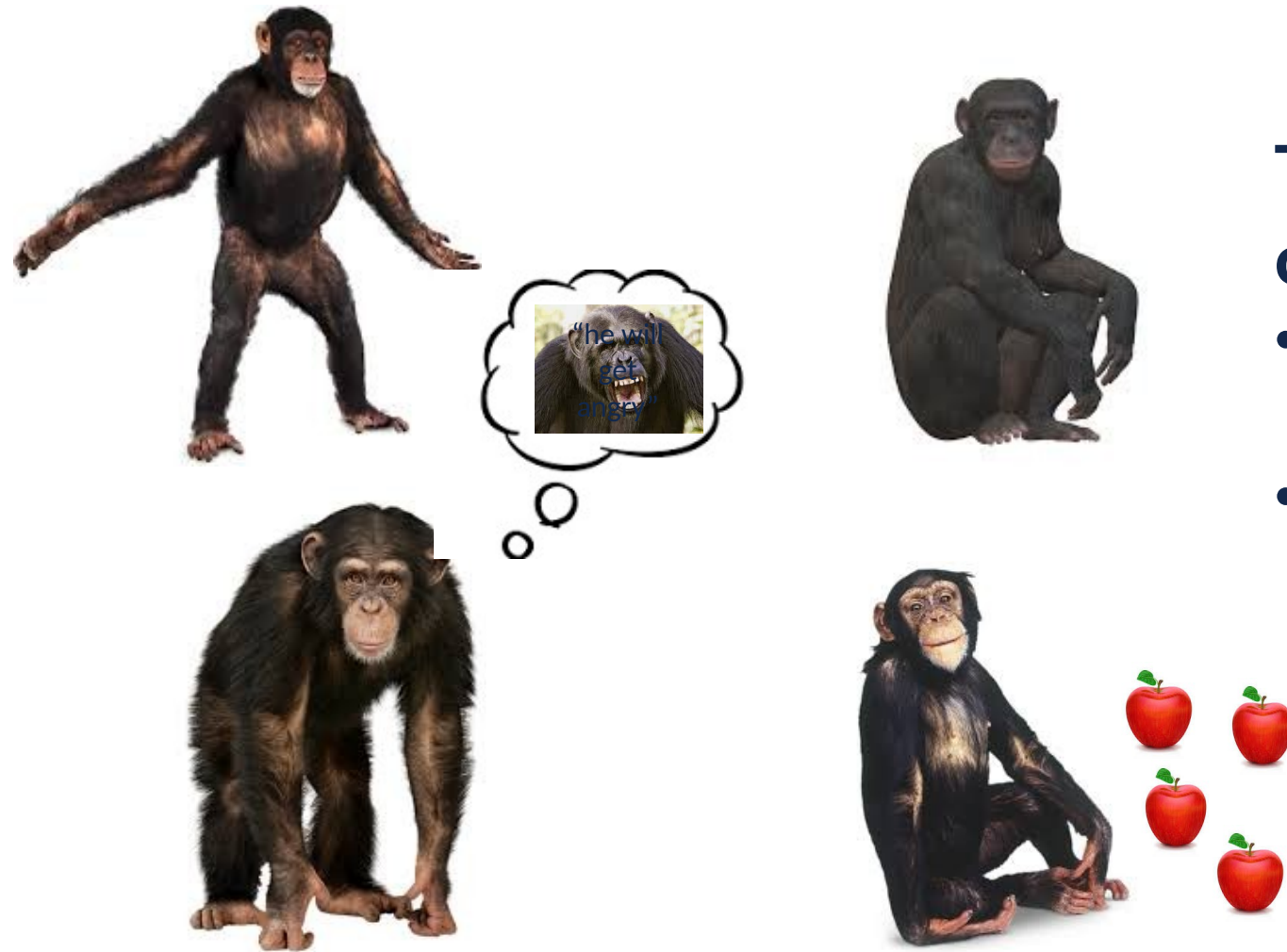


Fairness Responses

Presupposed Expectations About Others' Expectations

- Behavior changed when outcomes violated anticipated treatment
- Responses reflected deviation from a predicted norm, not raw amount
- Inequity responses implied models of what others “should” have done

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Two Compatible Interpretations of “Modeling Agents”

- **One view:** primates explicitly represent beliefs and intentions
- **Another view:** they predict others' future actions without mental content

Observed behavior is compatible with either interpretation

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Social Life Functioned as a Strategic Environment

- Outcomes depended on what other agents did, not just physical conditions
- Repeated interactions made future behavior alter current action
- Behavior reflected reasoning in a multi-agent strategic structure

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Long-Horizon Behavior Required Representing Distant Futures

- Primate actions often served outcomes not yet present
- Decisions reflected projected future states, not current rewards
- Behavior implies internal models of temporally distant consequences

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		C	D
	C	Take turns grooming	only A grooms B
	D	only B grooms A	no grooming

Deferred Reciprocity Implied Storing Conditional “Promises”

- Cooperative acts were given before knowing if they would be returned
- Behavior looked as if future obligations were represented internally
- Present action was guided by conditional expectations about later states

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Anticipation Extended to Personal Future States

- Behavior adjusted to secure the individual's own future welfare
- Agents behaved as if they projected their own later needs
- Long-term security shaped present choices even without cues

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Anticipation Occurred in a Strategic Multi-Agent Space

- Future payoffs depended on what others would do in response
- Representations had to incorporate others' anticipated reactions
- Social behavior reflected forecasting under uncertainty

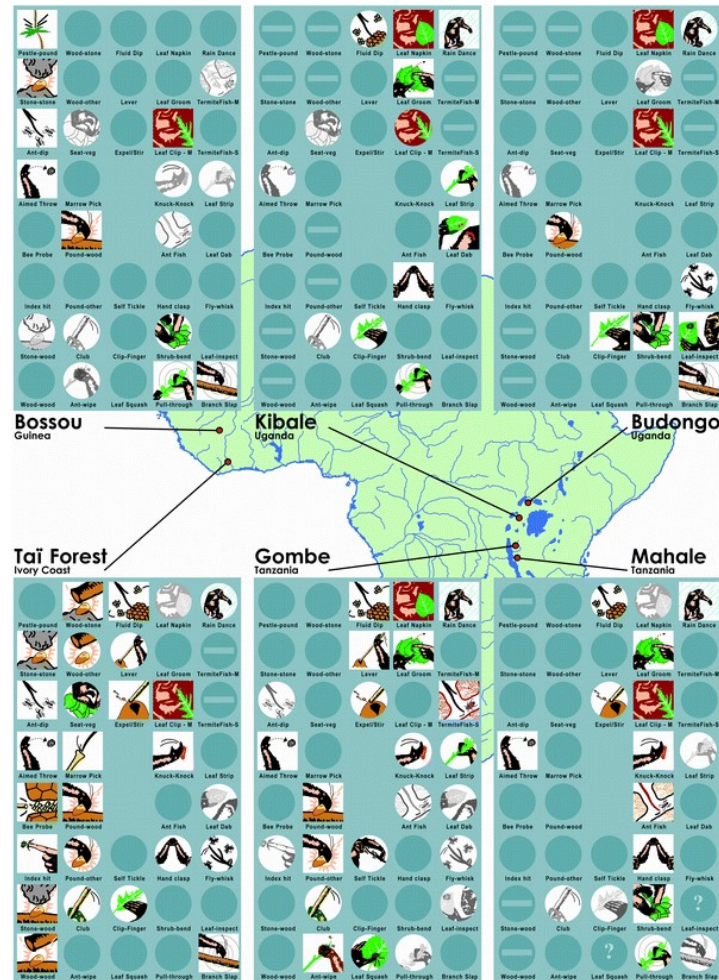
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Imitation and Copying

- Primates do not copy everything they see
- They preferentially copy competent or high-status models
- This implies internal evaluation of who is worth learning from

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Cultural Behaviors Required Learning From Other Agents

- Many primate skills were not individually rediscovered
- Behaviors spread through observation rather than trial-and-error
- Transmission implies extracting structure from others' actions

<https://www.nature.com/articles/21415>

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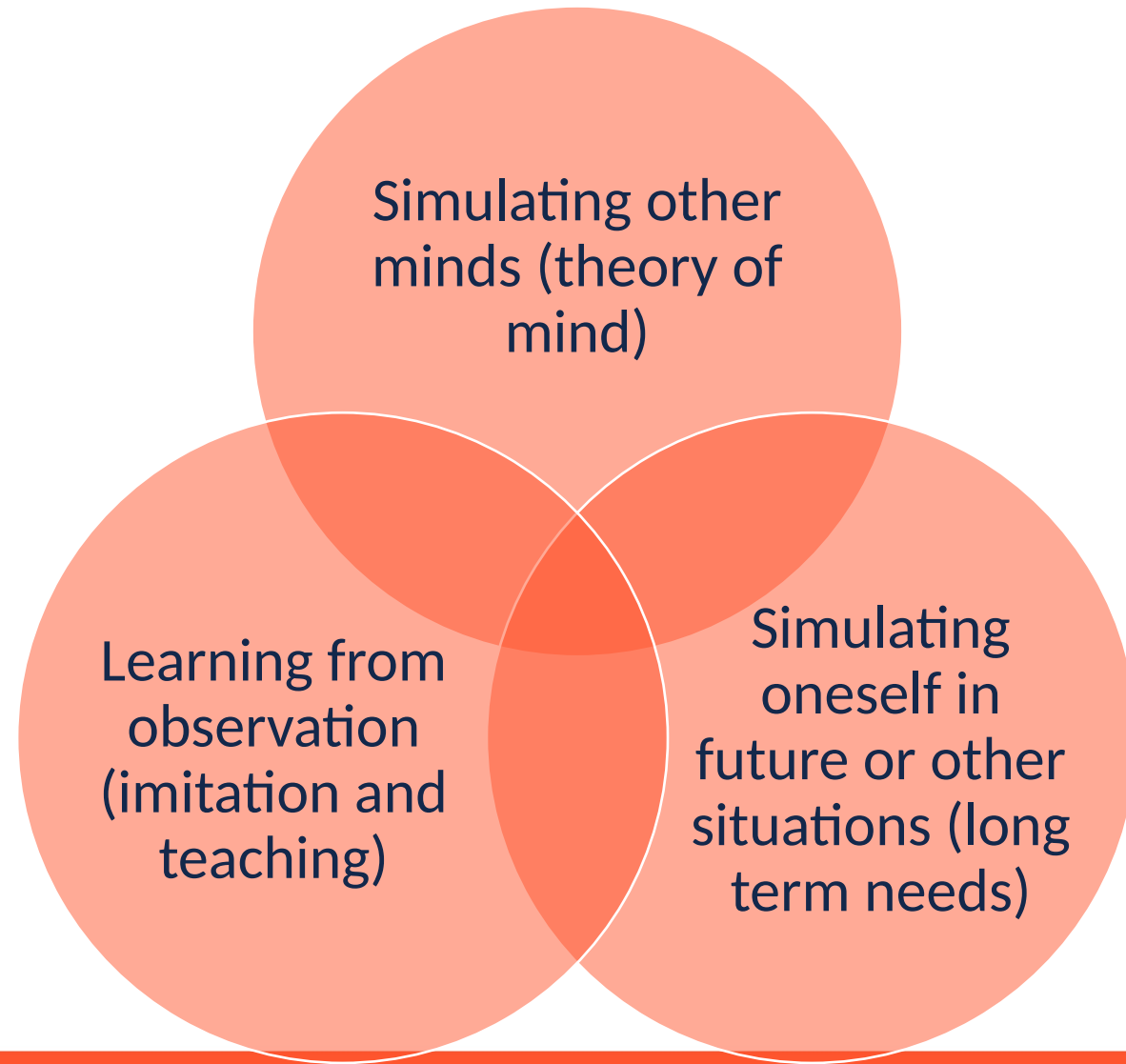


Culture Functions as Externalized Memory

- Group-level traditions stabilized information across time
- Individuals “outsourced” some knowledge to the social group
- This relieved cognitive load while extending behavioral sophistication

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**These Capacities
Jointly Explain Much
of Primate Behavioral
Uniqueness**

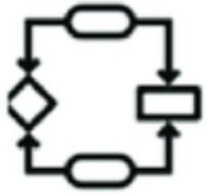


Primate Cognition: Algorithms & Representations



Computation

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Summary

- Capacities require representational storage and control
- Brain systems had to support long-horizon inference

Next Time

- Lecture 3 asks what neural changes made this possible